TITANIUM FORECASTING EVIDENCE

Model A and Model B performance review

One evidence page for two different tasks. Model A forecasts import prices and uncertainty intervals. Model B forecasts supplier-share movement probabilities. Their metrics are deliberately kept separate.

MODEL A VALIDATION

6 / 6 pass

All selected HS and horizon combinations pass validation gates.

MODEL A STRESS H3/H6

4 / 4 pass

Both products pass primary disruption horizons.

MODEL B VALIDATION

0 / 3 calibrated

All horizons beat both probability baselines.

MODEL B DISRUPTION H3/H6

2 / 2 pass

Primary horizons pass the required stress test.

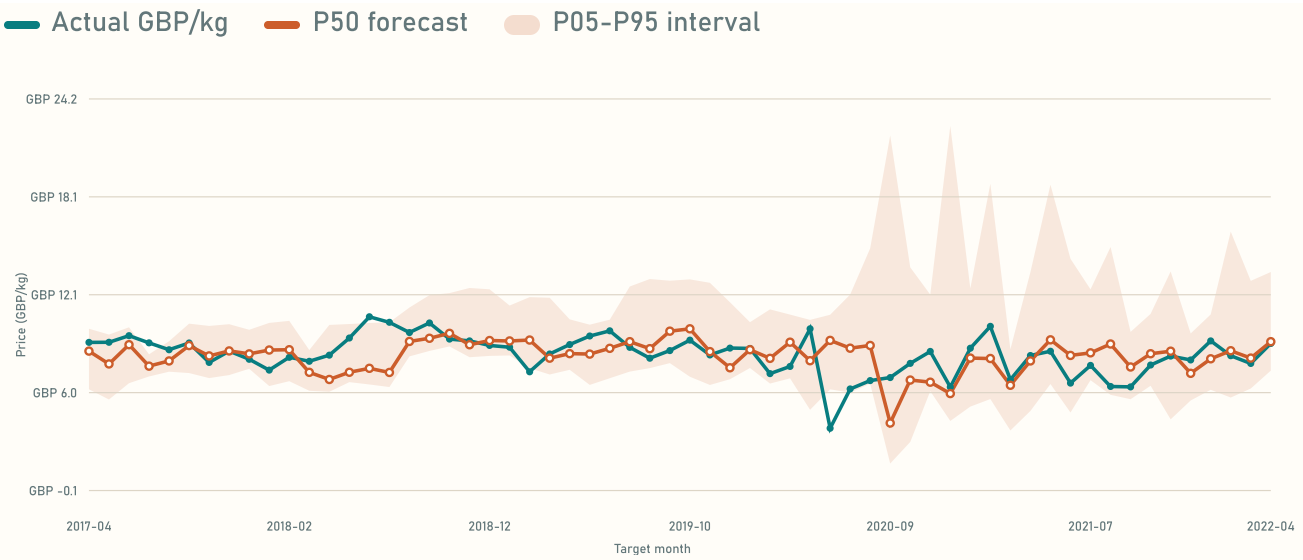
MODEL A EVIDENCE

Actual price versus forecast

P50 is the central estimate. The shaded P05-P95 range is the model's 90% uncertainty interval.

MAE	RMSE	SMAPE
GBP 1.08/kg	GBP 1.46/kg	13.7%

90% COVERAGE	MIS
93.3%	0.854 lower is better



Validation: linear_quantile passes. Its interval score is 97.0% of the no-change baseline.

GROUND TRUTH

Model A forecast records

ORIGIN	TARGET	ACTUAL	P05	P50	P95	COVERED	ABS. ERROR
2022-01	2022-04	9.028	7.343	9.140	13.464	YES	0.112
2021-12	2022-03	7.788	6.232	8.121	12.913	YES	0.333
2021-11	2022-02	8.268	5.672	8.578	15.969	YES	0.309
2021-10	2022-01	9.188	6.154	8.077	10.828	YES	1.111
2021-09	2021-12	8.004	5.490	7.176	9.650	YES	0.828
2021-08	2021-11	8.245	4.318	8.549	13.496	YES	0.305

ORIGIN	TARGET	ACTUAL	P05	P50	P95	COVERED	ABS. ERROR
2021-07	2021-10	7.691	6.403	8.391	10.865	YES	0.701
2021-06	2021-09	6.333	5.569	7.568	9.746	YES	1.236
2021-05	2021-08	6.360	5.842	8.996	15.011	YES	2.636
2021-04	2021-07	7.667	6.749	8.447	12.387	YES	0.780
2021-03	2021-06	6.567	4.744	8.286	14.282	YES	1.719
2021-02	2021-05	8.546	6.509	9.254	18.877	YES	0.708

MODEL B EVIDENCE

Actual supplier shift versus prediction

Four-class classification: collapse, fall, stable or rise. Accuracy alone is not sufficient, so probability scores and calibration are shown together.

ACCURACY	BALANCED ACCURACY	MACRO F1
45.7%	**40.9%**	**42.3%**

BRIER SCORE	MAX CALIBRATION GAP
0.643 lower is better	**25.7%**

Validation: The model beats both probability baselines and fails the 10% maximum calibration-gap rule.

CONFUSION MATRIX

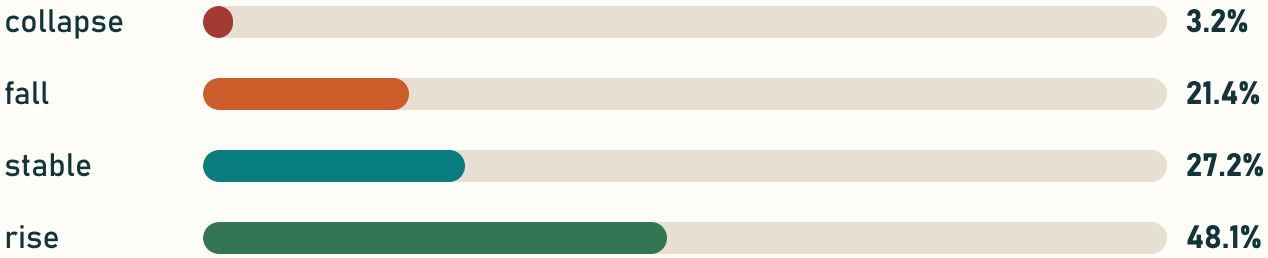
Ground truth against predicted class

Rows are actual outcomes; columns are model predictions.

	PRED. COLLAPSE	PRED. FALL	PRED. STABLE	PRED. RISE
ACTUAL COLLAPSE	3	10	0	1
ACTUAL FALL	2	37	17	16
ACTUAL STABLE	1	22	32	13
ACTUAL RISE	0	21	24	35

SELECTED RECORD

USA / 2021-06



Actual: **fall**. Predicted: **rise**. The prediction was incorrect.

GROUND TRUTH

Model B prediction records

ORIGIN	TARGET	SUPPLIER	ACTUAL	PREDICTED	CONFIDENCE	CORRECT
2021-03	2021-06	USA	fall	rise	48.1%	NO
2021-03	2021-06	Russian Federation	fall	stable	42.5%	NO
2021-03	2021-06	Other countries	rise	fall	51.4%	NO
2021-03	2021-06	Japan	fall	fall	31.7%	YES
2021-03	2021-06	France	collapse	fall	34.3%	NO
2021-03	2021-06	China	stable	stable	57.3%	YES
2021-02	2021-05	USA	rise	rise	45.8%	YES
2021-02	2021-05	Russian Federation	fall	stable	42.0%	NO
2021-02	2021-05	Other countries	stable	rise	47.8%	NO
2021-02	2021-05	Japan	stable	rise	51.6%	NO

ORIGIN	TARGET	SUPPLIER	ACTUAL	PREDICTED	CONFIDENCE	CORRECT
2021-02	2021-05	France	fall	fall	43.4%	YES
2021-02	2021-05	China	rise	stable	40.9%	NO

EXECUTIVE VIEW

What the evidence supports

These models cannot be ranked against each other because they predict different targets.

Model A: demo-ready research model

All six selected validation configurations meet their interval gates. The 2022-2023 H3/H6 stress tests also pass. Post-disruption performance is mixed, especially for H12 where incomplete future targets reduce coverage.

Model B: useful directional demo

Validation and test Brier scores beat both baselines, showing useful probability information. Exact probabilities need recalibration because worst-bin gaps exceed 0.10 in validation and post-disruption evaluation. H12 remains reporting-only.

SPLIT COMPARISON

Performance by evaluation period

Model A validation outcomes

HS	H	MODEL	COVERAGE	MIS RATIO	STATUS
810820	H3	linear_quantile	93.3%	0.970	PASS
810820	H6	adaptive_no_change	93.3%	0.978	PASS
810820	H12	random_forest_distribution	93.3%	0.774	PASS

HS	H	MODEL	COVERAGE	MIS RATIO	STATUS
810890	H3	elastic_net_residual	93.3%	0.869	PASS
810890	H6	linear_quantile	93.3%	0.796	PASS
810890	H12	elastic_net_residual	93.3%	0.800	PASS

Model B split outcomes

SPLIT	H	ACCURACY	BRIER	CAL. GAP	STATUS
Disruption test	H3	50.0%	0.603	7.7%	PASS
Disruption test	H6	47.2%	0.624	8.3%	PASS
Disruption test	H12	45.1%	0.621	19.8%	REVIEW
Post-disruption test	H3	56.1%	0.583	16.8%	REVIEW
Post-disruption test	H6	54.4%	0.585	47.5%	REVIEW
Post-disruption test	H12	58.3%	0.588	72.3%	REVIEW
Validation	H3	45.7%	0.643	25.7%	REVIEW
Validation	H6	45.8%	0.632	13.2%	REVIEW
Validation	H12	42.2%	0.647	28.2%	REVIEW

TRAINING DIAGNOSTIC

Why out-of-sample results come first

Model B apparent training scores are shown only to identify overfitting. Model A's saved package does not persist comparable in-sample predictions across its different quantile model families, so no artificial train-versus-validation curve is presented.

MODEL	HORIZON	TRAINING ACCURACY	TRAINING LOG LOSS	INTERPRETATION
Model B	H3	80.7%	0.754	Optimistic in-sample diagnostic; use validation and test metrics for decisions.
Model B	H6	79.7%	0.675	Optimistic in-sample diagnostic; use validation and test metrics for decisions.
Model B	H12	82.3%	0.800	Optimistic in-sample diagnostic; use validation and test metrics for decisions.

EVIDENCE DOWNLOADS

Auditable result files

Research evidence package. Sources remain limited to UN Comtrade, Bank of England FX and ONS ERUC/PPI. Lower MIS is better for Model A; lower Brier is better for Model B. These measures are not interchangeable.